

UNIVERSITY OF
Waterloo



Department of Mechanical and Mechatronics Engineering

Increasing Rocket Apogee by 23% Through Iterative Design

Changing 24 000ft to 31 000ft and Founding a Subsystem in the Process

A Report Prepared For:
The University of Waterloo

Prepared By:
Joel Godard

January 21, 2024

303 Westcourt Place
Waterloo, Ontario, N2L 6P5

January 21, 2024

Prof Bill Owen
Associate Chair, Mechanical Engineering
Department of Mechanical and Mechatronics Engineering
University of Waterloo
200 University Ave W.
Waterloo, Ontario, N2L 3G1

Dear Professor Owen,

This report, entitled Increasing Rocket Apogee by 23% Through Iterative Design, was written to fulfil my 3A work term report requirements. This is my 2nd work report submission. This report describes how an iterative design approach can have a significant impact on the results of student rocketry projects. The specific case study used is increasing the apogee for Waterloo Rocketry's 2023 competition rocket, Leviathan of the Sky. The intent of the report is to explain the reasoning, implementation, and results of this approach. Lessons learned throughout the project are called out to provide a resource for future rocketry team members.

This report was written entirely by me and has not received any previous academic credit at this or any other institution. I would like to acknowledge the help of the entire Waterloo Rocketry team which participated in changing the design approach. I would like to specifically thank multiple team members whose work overlapped with mine in major ways. Aaron Leszkowiat for his work on surface finish improvements and engine development, Christian Bicheru for his work on engine improvements. Roman Kobets for his work as the team's technical lead and for leading integration projects. Delainey Dymont for her work as a team lead and for her work on operations. Ryan Lau for his work on weather analysis and mass collection. Luca Scavone for helping to analyze LOTS flight data. And Shirley Kong for teaching me how to use OpenRocket and introducing me to rocket design. Each of these people, and many others I could not name contributed greatly to the larger project of ensuring a successful launch of LOTS and for this I am extremely grateful. Additionally, Waterloo Rocketry is sustained by the financial and technical donations of sponsors. I would also like to thank all the sponsors of Waterloo Rocketry for their contributions.

Best Regards

Joel Godard
3A Mechanical Engineering

Table of Contents

List of Figures	iii
List of Tables	iv
Summary	v
1.0 Introduction.....	1
1.1 Waterloo Rocketry	1
1.2 Hybrid Rockets.....	1
1.3 Student Rocketry Competitions	2
1.4 KOTS and LOTS.....	2
2.0 Problem Definition.....	4
2.1 Increase Apogee	4
2.2 Accurately Predict Apogee.....	4
2.3 Improve Timeline Adherence.....	4
2.4 Improve Integration.....	5
3.0 Iterative Design Cycle.....	6
3.1 Design Data Tracking.....	7
3.2 Engine Data Tracking.....	9
3.3 Simulation in OpenRocket	9
3.4 Loads Analysis	10
3.5 Mini Reports.....	11
3.6 Frequency of Cycles and Goals.....	11
3.7 Increasing Resolution of Measurements	12
4.0 Design Changes and Studies.....	14
4.1 Considering Diameter Changes.....	14
4.2 Engine Thrust Optimization	14
4.3 Fin Shape Optimization.....	15
4.4 Mass Distribution	16
4.5 Surface Finish Improvements.....	17
4.6 Antenna Study	18
4.7 Overall Length Reduction	18
4.8 Overall Weight Reduction.....	19
4.9 7075 Oxidizer Tank.....	19
4.10 Apogee vs. Wind Speed Designed For.....	20

4.11 Body Tube Changes	21
5.0 Results.....	23
5.1 Tuning Apogee.....	23
5.2 Achieved Apogee	24
5.3 Wind at Competition	24
5.4 Awards	26
5.5 Founding of Flight Dynamics	26
6.0 Conclusions.....	27
6.1 Reflecting on LOTS	27
6.2 Current/Future Impact of Iterative Design	27
6.3 Suggestions for Future Development.....	27
References	29
Appendix 1: LOTS Mass from Conceptual Design to Flight	30
Appendix 2: LOTS Predicted Apogee from Conceptual Design to Flight	35
Appendix 3: LOTS Flight Statistics.....	36

List of Figures

Figure 1: Hybrid Rocket Diagram [1].....	1
Figure 2: KOTS.....	2
Figure 3: Previous Design Approach.....	6
Figure 4: New Design Approach	7
Figure 5: Example Mass/CG/Length Tracking Sheet.....	8
Figure 6: Mass Breakdown by Section and Subsystem.....	9
Figure 7: Stability Free Body Diagram.....	16
Figure 8: Aluminum 3D Printed Bulkhead with Integrated Pneumatics Reservoir.....	19
Figure 9: Valve Cross Section	19
Figure 10: Boattail with Wall Thickness of 0.063in.....	19
Figure 11: SRAD Custom Pneumatically Actuated 1in Valve.....	19
Figure 12: Wind Speed vs Wind Speed Designed For	21
Figure 13: Image from Flight Video.....	26
Figure 14: LOTS Sections	30
Figure 15: Nosecone Mass Throughout Design.....	30
Figure 16: Recovery Mass Throughout Design.....	31
Figure 17: Payload Mass Throughout Design	31
Figure 18: Feedsystem Mass Throughout Design	32
Figure 19: Engine Bay Mass Throughout Design.....	32
Figure 20: Overall Rocket Wet Mass Throughout Design	33
Figure 21: Dry Rocket CG Throughout Design.....	33
Figure 22: Predicted Apogee Throughout Design Cycle.....	35

List of Tables

Table 1: Cycles Completed Along with Purpose	12
Table 2: Peak Thrust vs Apogee	15
Table 3: Surface Finish vs Apogee [9] (Units Converted)	17
Table 4: Wind Speed Data	20
Table 5: LOTS Flight Statistics	36

Summary

This report discusses how the apogee for Waterloo Rocketry's 2023 SAC rocket was increased from an early design estimate of 24 000ft to 31 000ft. It focuses on the creation and implementation of an iterative design approach used to identify apogee improvement opportunities, focus design energy, and track progress.

The main problems tackled in the design of LOTS were to increase apogee, accurately predict apogee, improve timeline adherence, and improve integration. This was accomplished by iteratively collecting design and engine performance data, running simulations, completing loads analysis, and reporting to the team on results.

Significant design changes were considered due to this system. Many of these were considered but not significantly acted upon including increasing rocket diameter, lowering design windspeeds, and changing oxidizer tank materials. Other design changes were acted upon in major ways. Wet mass was reduced by 18%. Rocket length was reduced by 10%. Engine peak thrust was increased by more 150% without increasing impulse. Body tube manufacturing defects were not accepted due to weight increase being allowable. The rocket surface finish was dramatically improved. Each of these changes were the result of significant study.

These design improvements led to the capability to overshoot the target apogee of 30 000ft. Therefore, focus was shifted to accuracy of simulation results and apogee was tuned down. LOTS launched in June 2024 to an apogee of 31 476ft. This result was achieved despite 98-99th percentile wind conditions making launch incredibly difficult. With this result the team won second place in the most complicated category at Spaceport America Cup. The project also established the Flight Dynamics subsystem as a major part of the Waterloo Rocketry team.

Concluding the report, reflections on the overall design process for LOTS are made along with discussing the current and future effects of this project and suggestions for future work.

1.0 Introduction

1.1 Waterloo Rocketry

Waterloo Rocketry is a student design team at the University of Waterloo that exists to provide students with opportunities to engage in hands-on learning through practical engineering work. Currently there are around 100 students involved with the team. Each year the team designs, builds and launches at least one rocket in an international rocketry competition; typically, the Spaceport America Cup. Waterloo Rocketry prides itself in designing and manufacturing as many components in-house as possible. The team has many subsystems to accomplish this. Selected subsystems include propulsion, airframe, recovery, software, and the focus of this report: flight dynamics and integration.

1.2 Hybrid Rockets

A rocket engine is an engine that uses Newton's third law to create propulsion. The basic principle of operation is that fuel and oxidizer are mixed and lit on fire. Then the combustion products are directed behind the rocket to create propulsive force. Most of Waterloo Rocketry's rockets have been hybrid rockets. This means that the fuel is a solid and the oxidizer is a liquid. This makes propulsion design significantly more challenging compared to simple solid rockets (where fuel and oxidizer are both solid) but offers more efficiency and integration opportunities. Figure 1 below shows the basic components of a hybrid rocket motor.

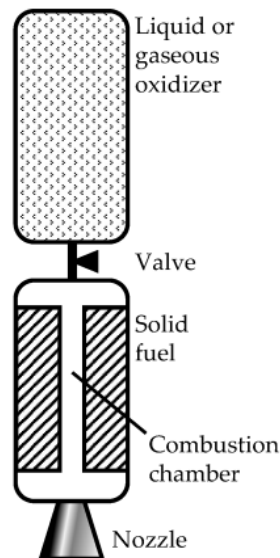


Figure 1: Hybrid Rocket Diagram [1]

For the purposes of this report, it is mainly important to note that the storage of a liquid oxidizer and the injection systems adds a lot of weight and complexity to a rocket of a student team size. This means that propulsion systems make up a large percentage of the overall rocket mass and that competing as a hybrid team is generally accepted to be more challenging than competing as a solid team.

1.3 Student Rocketry Competitions

There are many student rocketry competitions with Spaceport America Cup (SAC) being the largest in the world. The scoring for this competition (and most others) is mainly based upon technical aptitude and precision of apogee (max altitude). Points are awarded based on a technical report and how close teams are to various target apogees which can be selected. Waterloo Rocketry competed in the 30 000ft hybrid/liquid rocket category in 2023. This means a main goal of the team was to ensure the rocket flew as close to 30 000ft AGL as possible. Note that overshooting the intended altitude is penalized equally as undershooting.

1.4 KOTS and LOTS

Waterloo Rocketry's 2022 rocket was named Kraken of the Sky (referred to as KOTS) and due to a multitude of technical issues and competition failures was unable to launch. The rocket was 6in in diameter, 214in in length and 163lbs (wet mass) [2]. It can be seen in Figure 2:



Figure 2: KOTS

After the competition the team identified several notable problems that were desirable to address moving into the 2023 design cycle. Issues unrelated to this study are not discussed as there are many more.

- The most serious is that if KOTS were to have launched it would have only flown to around 24 000ft. As the goal of the competition is to fly to 30 000ft this would have a significant impact on scoring in the competition.
- Under high winds KOTS was relatively unstable due to low off-rail velocity of 92ft/s. This severely limits the wind cases which the rocket can be launched in and increases the risk of a failed flight. The main cause of this was poor integration and late-stage design changes.
- KOTS was completed on the drive to competition (with painting occurring in parking lots of hotels) leading to greatly increased team stress and design flaws which were unable to be fully fixed before attempted launch.
- Integration of various subsystems in KOTS was not fully considered due to a rushed development partially completed online leading to many unoptimized connections and components.

The above issues were mainly a result of increasing the complexity of KOTS by at least an order of magnitude compared to previous projects. This was chosen due to an increase in team members but created a multitude of issues to improve upon.

Leviathan of the Sky (LOTS) was the next rocket to be designed and this paper focuses on how the above issues with KOTS were corrected for LOTS. The intention in the design cycle of LOTS was to keep the overall architecture and engine configuration as KOTS but improve upon it. This was chosen to reduce project complexity and ensure LOTS would perform well. A secondary motivation for this approach is that redesigning every component every year is infeasible due to manpower limitations.

2.0 Problem Definition

The main goal for this project was to increase apogee. Secondary goals were to accurately predict apogee, improve timeline adherence and improve integration. These secondary goals were chosen based on competition scoring and as driving forces for increasing apogee.

2.1 Increase Apogee

The main goal set out at the beginning of the LOTS project was to increase the apogee to 30 000ft AGL. As mentioned previously, meeting this target would significantly improve competition scoring compared to KOTS. From first principles the easiest method to increase apogee is to increase the impulse the engine exerts on the rocket. Unfortunately, the engine for KOTS was at the limit of its class of rocket motor. This affects what is legally allowed to be built, transported across international borders, and insured for launches. Due to this, increasing the engine impulse was not a viable option.

2.2 Accurately Predict Apogee

While increasing the apogee compared to KOTS was the main goal; overshooting the apogee target is equally damaging to competition scoring. Thus, it is important to be able to predict the rocket apogee correctly and tune the system to achieve an apogee as close to 30 000ft as possible.

2.3 Improve Timeline Adherence

A major factor that enables increasing and correctly predicting apogee is timeline adherence. Historically the team has not been good at meeting the timelines which are set. This is a problem as a rocket is a very interdependent system and to optimize parts each designer needs time to consider the entire system. This time is required to optimize each part and ensure integration is done well. This challenge is compounded by the principle that integration opportunities are not easy to identify when projects are scrambling to keep up with timelines.

Some simple examples of this principle are:

- Painting must be consistent to create a predictable surface finish. Painting in a parking lot on the way to competition is not conducive to quality.
- Engine sizing, fin sizing and mass of each system impacts each other and all structural parts on the rocket which in turn affects fin sizing, engine sizing and mass of each system. This cyclicity means careful planning and good timing must be achieved to optimize how the systems interact.

Another major benefit of timeline adherence is that it dramatically reduces the stress level of team members. This in turn has a significantly positive impact on design quality and team happiness.

2.4 Improve Integration

The most practical improvement possible to increase apogee was to increase integration and apogee-focus while designing. As a result of integration: mass decreases, part count decreases, and designers are forced to consider the impact of their system on the wider rocket. This focus from each designer on how their part can be modified to increase apogee adds up to be significantly more than each individual component level changes on their own. This effect is very clear especially in aerospace applications such as rockets where weight and fuel mass are a critical concern. Reducing weight does not have a linear impact on apogee but instead an exponentially beneficial impact.

3.0 Iterative Design Cycle

The biggest change to the way the team functions throughout this project was implementing an iterative design cycle. Previously the team operated mainly by following the flow chart shown in Figure 3:

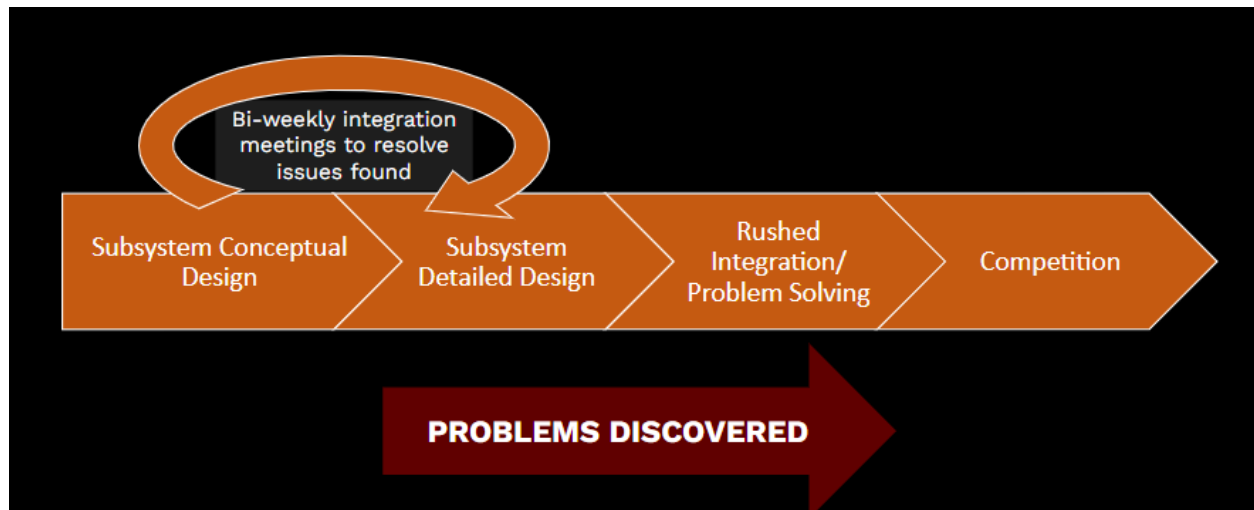


Figure 3: Previous Design Approach

Individually each subsystem would create a conceptual design of what they were considering. Next each subsystem would create a detailed design and work out interfaces with other subsystems through bi-weekly integration meetings. These meetings were intended mainly to fight major issues being discovered throughout the design process. As issues were found and competition grew closer the team entered into rushed integration and problem solving. Bringing many components designed in relative isolation together resulted in major issues being found very close to competition. An infamous team example is that while assembling the payload into the rocket a few weeks prior to competition it was found that the payload did not fit through the opening in the rocket. While this issue was resolved, it raises serious concerns about this approach and how robust it is for integration purposes.

Another note is that at no point in the process is there a natural point to look for integration improvement opportunities. Integration is used as a passive way to solve pressing problems, not proactively seek efficient design. This limits the amount of innovation the team can take on in order to improve.

To replace the above linear design approach, the method shown in Figure 4 was developed.

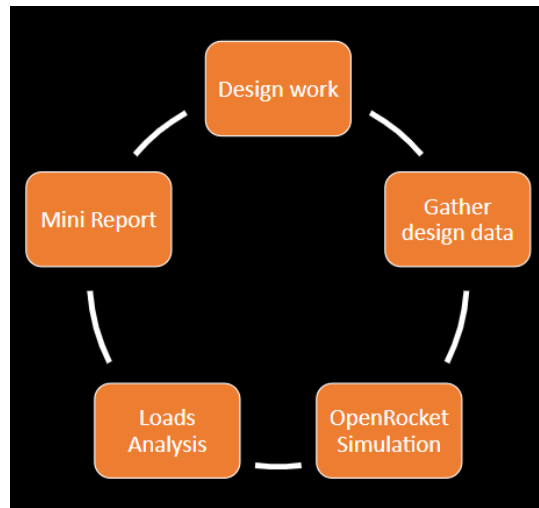


Figure 4: New Design Approach

In this approach design work is completed as before, individually by each designer. Then periodically this design data (mainly mass, and position data) is collected and put into a large spreadsheet. Based on this data OpenRocket simulations are done. This provides the key flight characteristics of the current rocket as it exists in each designer’s mind. Based on the outputs of the OpenRocket simulation, loads analysis is completed. This enables each designer to know what the limiting load scenario is for their parts. After each of these steps is complete, the relevant data is collected and presented in a 10–20-page mini report. This cycle was repeated throughout the design process at key project milestones to ensure the information remained recent enough to be representative. This also has a side benefit of driving design progress and keeping each subsystem aligned in terms of design progress.

3.1 Design Data Tracking

The main design data gathered in each cycle was:

- Minimum, expected, and maximum mass for each component.
- Expected CG location for each component (relative to the front of its section).
- The length of each section of the rocket and therefore the total length of the rocket
 - To improve communication the rocket was split into five segments. These were chosen at natural segment boundaries marked mainly by body tube couplers. This was done in the second cycle, so the earliest data does not include this segmentation.

- Initials of the person who entered the data, source of the data, and date entered for each data point above. This proved extremely useful for traceability.

An example of one of these spreadsheets is shown in full for scale below in Figure 5. To save space the size of this sheet has been reduced; for detailed numbers on mass data, reference the team technical report [3].

Figure 5: Example Mass/CG/Length Tracking Sheet

These sheets turned out to be very useful tools for team leads and integration teams to look through for opportunities for more optimization and just in general to get a picture of the current status of the rocket. One particularly useful output of this sheet is shown below in Figure 6. This shows the mass distribution of the rocket both by subsystem and by section of the rocket. This chart is extremely helpful to show that the subsystem with the largest potential for mass savings is propulsion. This allows the team to proportionally apply effort in the highest impact areas.

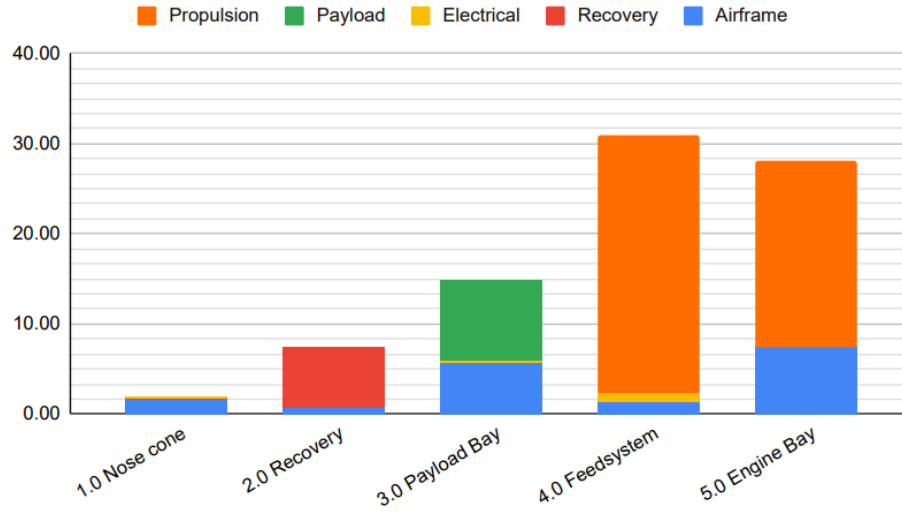


Figure 6: Mass Breakdown by Section and Subsystem

3.2 Engine Data Tracking

As the design for the engine progressed, the key engine data (thrust, mass and CG throughout time) was updated for each cycle. In the early stages this was mostly manually generated. A custom excel program written by the team was used to calculate center of gravity. This software is a modified version of what is discussed by Hilbert [4]. Similarly, a python simulation program was used for thrust data generation as discussed by Artem [5]. However as multiple static fires (full duration engine burn tests with the engine bolted to the ground) were completed and data was gathered, the later cycles used actual engine data from ground testing. Sometimes the engine design was in flux due to the results of static fire testing leading to design changes. In these cases, the data was modified using the same programs previously mentioned but using physical testing data as a base.

For each cycle, this key data was generated and synthesized by the propulsion team into an RSE file. This file can be read by OpenRocket and implemented in its simulations.

3.3 Simulation in OpenRocket

Using the spreadsheet mass data and the RSE file engine data, simulations were completed in OpenRocket for each cycle. This is an opensource software commonly used for simulating rocket flight. It uses a modified version of the Barrowman equations to calculate the approximate flight performance of model rockets [6]. This approach simplifies the aerodynamics of the rocket to

avoid resorting to more complicated or computationally expensive simulation methods. Methods such as CFD, wind tunnels, and full-scale testing would be more representative. However, while OpenRocket is not the most rigorous way to computationally model high velocity rocket dynamics, the majority of better tools are not available to the team. This is partially due to complexity and mainly because many high speed aerodynamic simulation software suites have significant defense technology applications and are therefore not publicly available.

The main metrics tracked varied with each cycle based on goals for the cycle but mainly were:

- Apogee (ft)
- Off-rail velocity (ft/s)
- Minimum stability in expected wind case (Stability Margin Calibers)
- Minimum stability in maximum allowable wind case (Stability Margin Calibers)
- Maximum stability (Stability Margin Calibers)
- Max Acceleration (both G and ft/s^2)

For each cycle, general simulations were done to compare the rocket to the previous cycle's data and get a sense of the current state of the rocket. However, there were also specific studies that were carried out using OpenRocket to compare design concepts against each other. Some notable studies included varying surface finish of the paint, changing material for the oxidizer tank, and a study on antenna size. These are each discussed later in the report. But it is important to note these improvements came out of specific studies in OpenRocket. This was used to show where limited design effort and time was best spent to gain the most apogee possible. Some of these studies were also completed outside of any specific cycle as the model was available and easy to use.

3.4 Loads Analysis

In cycles where loads analysis was especially relevant it was completed using the outputs of OpenRocket as inputs to the loads analysis software. Loads analysis is outside the scope of this report. However, it is sufficient to know that it allowed the team to understand the loads on each individual part in the rocket throughout flight and pinpoint the limiting load case. This was not required to be precise near the beginning of the design cycle but was important to compute for detailed design work in later stages.

3.5 Mini Reports

For each cycle the team ran, there were significant learnings both from general investigation and specific concept studies. This necessitated a way to communicate the learnings to the rest of the 100-person team. For some of the cycles this was done by writing a mini report (around 10-15 pages) showcasing the simulations done and the main learnings that were applicable to the average team member. Included was also a one-page summary of the most important information. As the design cycle moved along it was noticed that very few members read much more than the one-page summary and the report took a significant amount of time to write. This time could be better spent running further simulations. Therefore, in the later cycle's learnings were distributed in more of a one-page fashion. And when more detailed information was required it was communicated using Slack messages to specific team members. If more time and energy were dedicated to the project, full mini reports would have been written for each cycle, but time and energy are often very limited resources on a student team.

3.6 Frequency of Cycles and Goals

A key consideration of the project was how many cycles to complete and when to do them. The rationale was that a cycle should be completed when:

- New data is available that significantly changes the flight profile (i.e. completing a static fire test)
- A major architecture design needs to be made and data is required to make that decision (i.e. fin sizing or thrust requirement definition)
- A major competition deliverable is required and updated numbers are desired (i.e. competition final design report)

The cycles completed are shown below along with their purpose in Table 1.

Table 1: Cycles Completed Along with Purpose

Cycle	Date	Purpose
Pre-Cycle 1 (cycle 0)	August 2022	What do we want to change from last year's rocket? What thrust should the engine have?
Cycle 1	November 2022	What is each subsystems conceptual design and how can we optimize it as early as possible where easy opportunities exist?
Cycle 2/2.1/2.2/2.3	February 2023	Updating inputs based on static fire and completing fin sizing.
Cycle 3	May 2023	Updating inputs based on detailed designs being "complete" and generating numbers used for final competition design report.
Cycle 4/4.1	May 2023	Updating inputs based on more fabricated part masses being available. Cycle 4.1 was kept updated until rocket construction was done to obtain a log of individual part weights.
Cycle 5/5.1	June 2023	Switching to weighing assemblies instead of parts for greater accuracy. Results used to determine ullage in tank (last opportunity to tune engine performance)
Cycle 6	June 2023	Only using as-fabricated masses with paint etc. accounted for day of weather and used for apogee prediction at competition. Used to set wind limits for launch.

As the pace of manufacturing increased near the end of the project the pace of cycles also increased. This was due to the increase in new data as parts could be weighed "as built". Also, it is important to note that while simulation work mainly occurred in the design phase, cycle 6 continued until the day of launch. This is when a final apogee prediction was made based on day of launch weather conditions and given to the competition judges. This prediction is used for computing some of the additional awards at competition.

3.7 Increasing Resolution of Measurements

Compared to the design and simulation of KOTS significant effort was put into increasing the resolution and confidence of the measurements for the mass of the rocket. For example, in KOTS many components were not fully weighed and items such as paint were missed entirely. Following along with the focus on increasing the accuracy of simulations, it was thus necessary to keep track of the mass of every component possible. Prior to manufacturing of components, SolidWorks was

used to compute the mass of most components. Previous experience with more exotic materials such as composites was also used for non-homogeneous materials. Then as components became available, they were weighed individually. As each assembly became available it was weighed to ensure screws, lubricating/sealing fluids, paint and other miscellaneous items were included. This data was then used for the final simulations. As a side note, mass data was asked for so many times that inside jokes were created about mass collection.

4.0 Design Changes and Studies

4.1 Considering Diameter Changes

Aspect ratio refers to the length of the rocket divided by its diameter. This has significant impacts on stability, loading and other flight characteristics. Early on it was realized that KOTS had a very high aspect ratio compared to most rockets (35.67). For reference, typical tactical missile aspect ratios range from 8-18 [7]. This high aspect ratio can be an issue due to increasing structural loading. Note that military rockets often have low aspect ratios due to high lateral acceleration/high angle maneuvers. For KOTS/LOTS these high loads could be remedied by designing strong structures. However, despite this, a study was done to determine if increasing rocket diameter and decreasing length would increase apogee. It was initially expected that this would increase apogee but a major factor in this decision is that the team has previously significantly invested in 6in hardware (tanks, moulds, etc.). After completing this investigation, it was found that after switching to a larger diameter:

- Fin size was required to significantly increase due to the CG moving farther back.
- This fin size change dramatically increased weight and drag.
- That impact along with other unknown aerodynamic factors made apogee significantly lower for an 8in diameter rocket, against initial expectations.

Using this result the diameter of LOTS was left at 6in. This is a good example of how rocket optimization problems are usually not simple enough to intuitively guess the result. This is why robust simulation with good inputs is required to separate feelings from facts.

4.2 Engine Thrust Optimization

Pre-cycle 1 (cycle 0), the key output was to set the desired engine thrust level. As mentioned previously, the impulse of LOTS engine is locked by regulations. However, this impulse can be released all at once in a higher thrust shorter burn or released slowly in a lower thrust longer burn. Some other considerations:

- A higher peak thrust leads to significantly increased stability off the launch rail due to faster acceleration and thus a higher speed when leaving the rail.
- A higher peak thrust lowers thermal requirements on the engine as the system can be designed in such a way that heat will not soak into heat sensitive components until after the burn is over.

- There are significant structural considerations to increasing the peak acceleration of liftoff beyond the peak acceleration of parachute deployment.
- Higher peak thrust increases the max speed of the rocket bringing it definitively into the supersonic regime, which is harder to model aerodynamically and increases the risk of unknown supersonic effects.

This study was done by taking the KOTS mass values and resizing engine files. Simulated apogee values are shown in Table 2.

Table 2: Peak Thrust vs Apogee

Peak Thrust as % of KOTS Engine	Apogee (ft)	Off Rail Stability (Cal) – Target minimum 1.5
50%	16 770	0.86
80%	23 450	1.20
100%	24 200	1.55
120%	24 410	1.90
150%	24 450	2.12

Based on the results of this study the engine thrust was set at around 150% of the KOTS engine design (10.9kN peak thrust). This was chosen to maximize the apogee and increase stability while comparatively limiting the structural implications of an extremely high thrust. For perspective on the forces involved, this resulted in a 0-60mph time of 0.22 seconds with peak vertical acceleration of around 16G's.

4.3 Fin Shape Optimization

Rocket fin design is a complicated design challenge. For the purposes of this report, it is mainly relevant to note that good data input and confidence is required to lock fin size early in the design cycle. Locking fin size early is a major goal as making the fin can requires 250+ person hours. The earlier this work can be started, the better. Due to the cycle-based design system, fins were able to be locked in late February which is around a full month earlier than during the design of KOTS. For more details on fin design, reference my previous paper on that project [8]. Many optimizations were made in the fin shape specifically designed for the flight profile of LOTS. This resulted in around a 3 000ft apogee improvement compared to using the same fins as KOTS.

4.4 Mass Distribution

As mentioned in the previous section, reducing the drag and mass of the fins can have a huge impact on maximum apogee. The purpose of the fins is to ensure the rocket remains upright through passive stability. Knowing this, the question can be raised: is there any other way to increase stability besides fins? Stability is mainly determined by the distance between the center of gravity and the center of pressure of the rocket. This can be visualized by thinking of the rocket as a see saw with the pivot at the center of gravity. When it pivots around the center of gravity the force from the air (acting at the center of pressure) can rotate it back into place if the center of pressure is behind the center of gravity. This effect is shown in Figure 7.

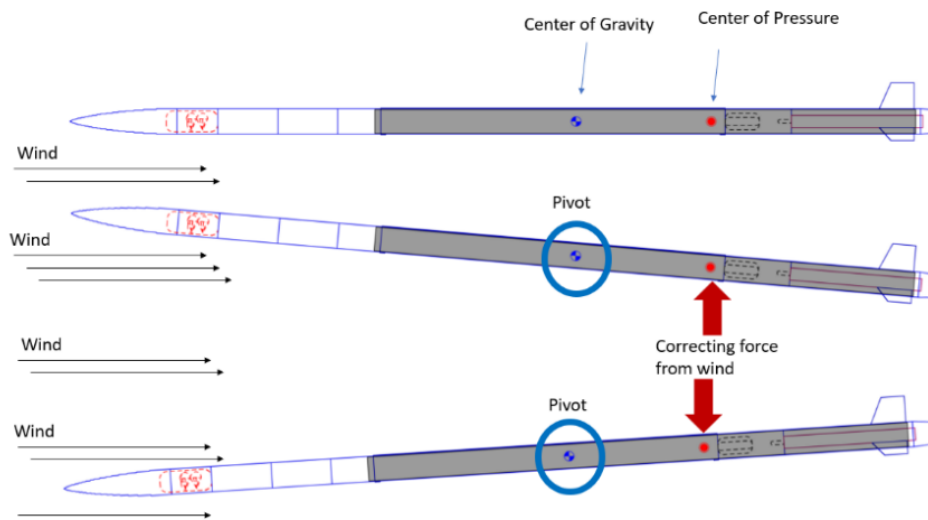


Figure 7: Stability Free Body Diagram

Using this understanding, if the center of gravity is moved forwards, stability is improved. Therefore, it is most critical to reduce weight near the bottom of the rocket and weight at the top of the rocket is less relevant. In fact, for certain rockets adding weight in the nosecone and reducing the fin size can increase apogee. While this was explored, it was not worth adding any significant weight to LOTS. This knowledge was used to focus weight reduction efforts on the bottom section of the rocket. When possible, components were moved further up the rocket. This enabled a reduction in the size of the fins and therefore an increase in apogee. Note that LOTS high aspect ratio also increases passive stability.

4.5 Surface Finish Improvements

Surface finish can have a significant impact on rocket apogee. After studying this it makes intuitive sense, but this was not a major focus at the beginning of the project. It was discovered around cycle 2. Interestingly, when exploring improving apogee, paint was by far the best effort to reward ratio of any project. Technical details are beyond the scope of this report but fortunately another team member wrote a report on this topic entitled *A Practical Method For Reducing Skin Friction Drag in Sounding Rockets* [9]. Table 3 shows data from this report. Note that these values were calculated using a simulation file with incorrect assumptions leading to the apogee numbers being unrealistically high. However, the relative difference between the surface finish options is believed to be representative. Units have been converted to feet to match the rest of this report.

Table 3: Surface Finish vs Apogee [9] (Units Converted)

Surface Finish Option	Apogee (ft)
Rough	27 205
Rough unfinished	28 789
Unfinished	29 941
Regular paint	31 946
Smooth paint	34 219
Optimum paint	35 531
Aircraft sheet-metal	35 535
Finished/polished surface	35 535
Mirror	35 535

After discovering this effect, a major effort was made to characterize the surface roughness of the rocket using a surface profilometer. Simultaneously work was done to improve the surface roughness to a much smoother level. Contributing around 3 000 – 4 000ft to the apogee improvement, this change alone was a significant factor in reaching apogee goals. It also increased the team’s ability to predict the rockets apogee due to greater understanding of the inputs used in simulation.

It is important to note that the team was unable to remove every source of surface imperfection. Protruding bolts, coupling interfaces, and manufacturing defects increase drag and thus

engineering judgement was used to correlate surface profilometer measurements of ideal samples to the rocket as a whole. This is also discussed in the report mentioned above.

4.6 Antenna Study

In the custom student design of a new antenna solution for live telemetry the question was raised, how much does the length of the rocket taken up by the antenna matter? The space allocated can influence RF reception significantly due to interference, but it was uncertain how much impact this would have on apogee. A study was completed into how much apogee would be reduced due to this additional length (requiring additional structural length). This study found that the additional length recommended for improved RF reduced apogee by around 50ft. As this is extremely minimal, it was determined to add around 6in of mostly empty space to improve RF performance and ease of assembly. This example proves the value of robust flight simulation to ensure design decisions are made based on data instead of intuition.

4.7 Overall Length Reduction

While in the previous study, length reduction was not desirable, due to improvements in integration, engine efficiency and other factors, the overall length of the rocket was reduced by 38 inches (8%). This happened slowly throughout the development cycle so was never deeply studied in terms of apogee gain. In theory it should increase apogee by reducing wetted area, but this is not an extremely significant gain. Very roughly assuming 50ft gained per 6in reduced (matching the antenna study) the 38in reduction increased apogee by roughly 316ft. However, length reduction was mainly made in the oxidizer tank. This results in much more significant weight impact as the oxidizer tank has a much higher linear weight than other areas of the rocket. This effect is captured in the results of general weight reduction efforts.

Length reduction also had the minor side benefit of bringing LOTS closer to an “ideal” aspect ratio.

4.8 Overall Weight Reduction

Weight reduction on an aerospace vehicle is paramount to increasing apogee. This is because reducing weight decreases the amount of fuel needed for the same acceleration by a nonlinear amount. The largest weight reduction on LOTS was achieved by major engine efficiency gains which achieved the same impulse with significantly less fuel. However, there were also numerous weight reduction measures that added up to major savings. Small and major changes such as thinning couplers to the minimum machinable value, adding holes and notches to non-structural components, custom valve design, and pursuing exotic materials had a major impact.



Figure 8: Aluminum 3D Printed Bulkhead with Integrated Pneumatics Reservoir



Figure 11: SRAD Custom Pneumatically Actuated 1in Valve

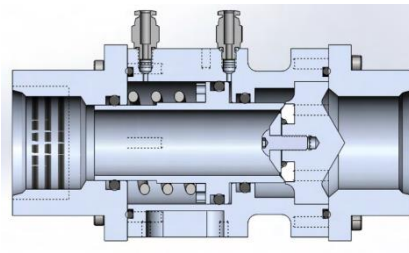


Figure 9: Valve Cross Section



Figure 10: Boattail with Wall Thickness of 0.063in

The weight reduction efforts throughout the LOTS program achieved an 18.7% weight reduction from KOTS mass (27lbs).

This was possible through a significant entire team effort on reducing mass. One good example of how mass conscious the designers of LOTS were is the payload weight. It weighed in at exactly the minimum mass required from the competition, 4.00Kg (around 8.82lbs).

4.9 7075 Oxidizer Tank

One easy weight reduction target is the oxidizer tank. As this tank weighs 20lbs it makes up around 1/4th of the final LOTS dry mass. The main component is an aluminum 6061 tube which forms the outer surface of the rocket and also acts as a pressure vessel. The team explored replacing this with

an aluminum 7075 tank which would reduce weight by around 10lbs. However, the cost of replacing this component was found to be around two to three thousand dollars. This was prohibitive for the team and due to many other improvements throughout the project, replacing the 6061 with 7075 was abandoned. In hindsight some of the other components which were changed from 6061 to 7075 (injector valve etc.) showed major corrosion problems when exposed to nitrous oxide. This would have caused major operational issues trying to maintain a corrosion-free surface on the inside of the tank.

4.10 Apogee vs. Wind Speed Designed For

As previously mentioned, the size of the fins has a dramatic impact on drag and therefore to optimize performance the fins should be as small as possible. However, increasing the size of the fins increases stability. Increasing the stability of the rocket increases the amount of weather conditions the rocket can launch in as well. This presents a design challenge as ideally the team would be able to launch in all wind conditions feasible at the launch site. Designing this way would limit apogee so another way was needed. Note that the competition does not allow teams to launch at higher than 20mph wind so this could be an upper bound. However, to consider if the rocket could be designed to an even lower wind value, analysis of the wind conditions at the launch site was completed. Weather conditions were taken from Visual Crossing [10] for the week of competition (only considering the hours with an open launch window) in the last three years and analyzed to generate Table 4.

Table 4: Wind Speed Data

Wind Speeds (mph)	<=5	>5 and <=10	>10<=15	>20
Percent of Time	22.73%	43.94%	28.79%	0.91%

Looking at Table 4 it appears extremely unlikely that the wind at the launch site would be greater than 15mph when launching. And it is also noticeable that even with lower wind limits, there could be a good number of launch opportunities.

To confirm how much altitude was on the table, further study was done to consider how much apogee would be lost by designing the fins for higher wind speeds. Simulations were done to define how large the fins would need to be in order to have equal stability at each of the wind speeds

shown below. Then these fin sizes were each simulated under the same conditions to compare the maximum apogee. The results of this study are shown below in Figure 12.

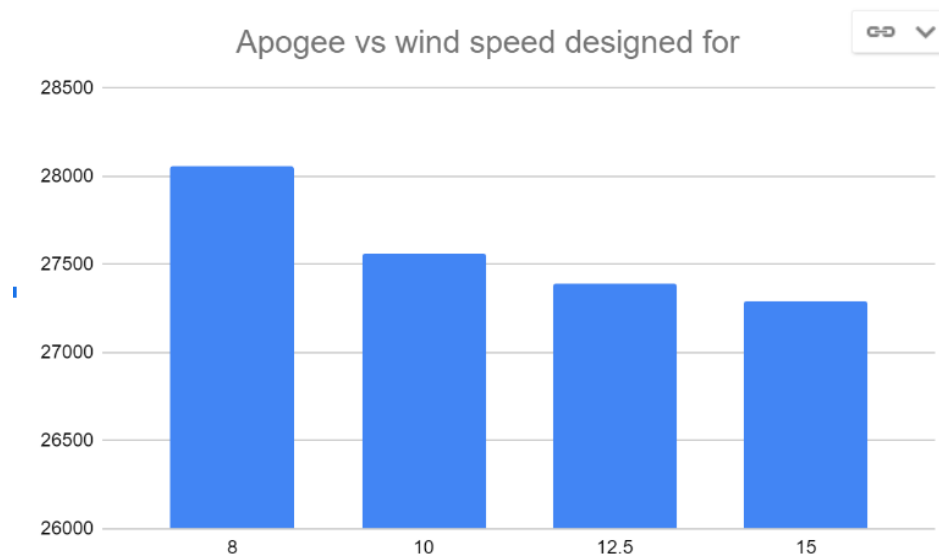


Figure 12: Wind Speed vs Wind Speed Designed For

From this data, if the rocket was designed to be only capable of launching in 8mph wind it would have 750ft more apogee than if it was designed to launch in 15mph wind. While this is significant, the risk of being unable to launch was considered not worth such a small altitude gain. Thus, the rocket was designed to be launchable with satisfactory stability in 15mph wind. Note this is less than when the competition closes the range (20mph). At this speed they consider it too dangerous to launch any rockets.

4.11 Body Tube Changes

While the majority of LOTS mass maximum and minimum predictions were accurate, one notable exception is worth describing. When manufacturing student researched and designed (SRAD) composite body tubes were kicked off, multiple quality issues were discovered. This led to significant doubt about the load bearing capacity of these parts. Due to simulations completed at the time suggesting around a 30 000ft apogee it was determined to consider the effects of switching to Commercial off the Shelf (COTS) body tubes in key areas. After simulating the apogee loss, it was deemed worth it for overall project schedule to abandon SRAD body tubes in specific locations in order to reduce manufacturing strain and improve part quality in other SRAD composite parts. Another side benefit of this change was an increase in stability due to the increase

in weight above the previous center of gravity. Without timely simulation data to prove that SRAD parts were not required to reach apogee goals, the resulting greater than one pound mass increase would not have been accepted.

5.0 Results

5.1 Tuning Apogee

As the team made significant progress in weight and drag reduction, engine efficiency, and surface finish it became clear, around the end of cycle 3, that 30 000ft was achievable. After this point the focus shifted to accurately bringing apogee as close to 30 000ft as possible. As the number of interconnected parameters possible to shift was immense it was not trivial to solve for what was the best way to predictably control apogee. The ideal lever to control apogee would be linear, predictable, and easy to implement as close as possible to launch without affecting other aspects of flight. Varying the amount of oxidizer loaded onto the rocket excellently matches these criteria. This can be accomplished in a few ways. The most obvious and best would be to weigh the rocket on the pad and measure the weight difference while loading oxidizer. However, carrying this out in practice while the rocket is partially supported by the launch rail proved to be extremely difficult. So, this approach was abandoned. Instead, using a dip tube leading to a vent valve, the minimum ullage of the oxidizer tank can be set. This allows the tank volume to be accurately controlled. However, this dip tube was required to be installed into the tank at the latest a week before travelling to the competition to ensure proper sanitation. Due to this manufacturing constraint weather conditions and other day-of effects cannot be accounted for. However, it proved to be sufficient in this case.

To determine how long the dip tube should be, first the weather conditions must be estimated. This was done using the previously mentioned weather analysis and assuming the team would launch in early afternoon in around average wind/temperature/pressure conditions. Then all mass and dimension data fed into OpenRocket received a comprehensive review to ensure that every bolt, lubricating fluid and dimension change due to paint thickness was accounted for. After this, engineering judgement was used to set the surface roughness of the rocket based on surface profilometer data to account for exposed bolt heads, coupling sections, other imperfections and known simulation inaccuracy. For more details on this see [9]. Then custom engine file generation software was used to iteratively bring the apogee result from OpenRocket to exactly 30 000ft. This resulted in a nominal oxidizer mass of 15.63kg.

Other solutions for controlling apogee that were considered but not implemented included:

- Adding a controlled amount of weight to the rocket on launch day. This was not implemented due to the difficulty of mounting unplanned mass in the rocket without upsetting mass distribution/stability and affecting loads in an unknown way.
- Decreasing the surface finish of the rocket. While the surface finish achieved was not strictly required for reaching 30 000ft, accurately characterizing the surface finish achieved is required to predict apogee well. As lab testing requires significant effort, samples of a lesser surface finish were not analyzed. Due to this, reducing the surface finish of the rocket was rejected. The effects of this reduction would have been too difficult to analyze in the timeframe allocated.

5.2 Achieved Apogee

LOTS flew to an apogee of 31 476ft AGL based on altimeter data. The flight simulation performed on the day of launch considering day-of weather was 30 380ft making the percent error only 3.7%! As far as the team is aware, this is the closest prediction for apogee from a 30 000ft hybrid student team at SAC ever. Goals for mass reduction and apogee improvement were well overshoot. It is estimated that through implementing all improvements considered, it would be possible to achieve an apogee of 40 000ft without increasing engine impulse.

5.3 Wind at Competition

While previously stated that launching in above 15mph wind would be an eventful occurrence, the wind speeds seen at SAC in 2023 were extreme. Values recorded were around the 98-99th percentile of previous wind conditions in the last three years. This was realized around a week before the competition through looking at forecasting. To ensure that launch was still possible, the following steps were taken:

- An anemometer was procured, and software was written to read out wind speed live from the rocket pad to mission control during launch operations.
- Significant work was done to increase the launchable conditions through deeper research on stability and it was concluded that (with the latest mass data) the rocket could launch with acceptable risk in around a 22mph wind gust. Note that wind gusts and sustained values are significantly different but not in the scope of this report.
- Research was done on professional weather prediction tools, and it was found that, during the competition, at around 1pm each day the wind would significantly pick up to around 20-28mph before picking up even more later in the day.

- Rocket and launch pad setup operations were extremely optimized. Procedures were written and members were assigned for every task required before flight to enable launching earlier in the day, while ensuring no steps were missed.
- Launch procedures were modified to include a possible hold for wind conditions in case gusts came up unexpectedly. Similarly, the time from when ops was restarted to when the rocket would take off was defined to be around 10 seconds for delay in ignition.

Despite these efforts the rocket was not ready to start ops until around 12pm. Then as the wind was dramatically increasing from 5mph average to a range of 10-18mph the decision was made to proceed with ops while at risk of scrub. In the last minutes of fueling the rocket, the wind suddenly gusted to 28mph and stayed around this value. Fueling was complete but the decision was made to hold in hope of lower wind speeds. After waiting 3-5 minutes, the wind speed dropped back down to around 18mph. After waiting for 1-2 minutes to ensure this trend would continue, the decision was made to launch. LOTS launched at 12:42PM with 13mph wind recorded at the pad! Around 20 minutes after the LOTS launch, the range was closed for the entire day due to high wind.

No other hybrid teams were able to launch the first day of launches due to the wind and comparatively slow setup times. It was only due to the incredible optimization of operations and live flight dynamics support at mission control that the launch was not delayed to a future day or scrubbed entirely. Less than 80% of all teams in the competition were able to launch with many of these failures due to the wind. Of that 80%, many teams launched in very suboptimal conditions. After the team's launch Waterloo Rocketry was able to assist other teams and the competition with live wind data as no other teams had this capability.

These actions and the result would simply not be possible without the flight dynamics effort throughout the entire design cycle to improve understanding, capabilities, and integration among the team.

5.4 Awards

The team won second place in the 30 000ft hybrids/liquids category in a close race against PUT Rocketlab and won best flight footage in the entire competition shown in Figure 13.



Figure 13: Image from Flight Video

5.5 Founding of Flight Dynamics

Before the 2023 design cycle, flight dynamics was considered in a very minimal fashion. The airframe subsystem team members did basic OpenRocket simulations to size fins but almost no other flight dynamics work was completed. Throughout the course of this project, due to the scope of work increase, Flight Dynamics has been established as one of the main subsystems on the team. It is currently composed of more than 6 team members who meet bi-weekly to ensure flight dynamics tasks are proceeding in an effective manner. In the future, these team members will hopefully accomplish significantly more than was accomplished through this one project.

6.0 Conclusions

6.1 Reflecting on LOTS

LOTS is perhaps the greatest success in Waterloo Rocketry history. The main cause of this is the increase in team members and the commitment levels across all projects (with many names that could be specifically called out). However, despite this team growth it is hard to see how reaching 30 000ft would have been achieved without an iterative design approach. The impacts of this approach were far reaching. From obvious areas like propulsion design to less obvious impacts such as influencing recovery and payload design. The implementation of this system enabled the team to collaborate in a much more cohesive way, leading to cross subsystem integration in a way often lacking in student team projects. The lessons learned from this design cycle will hopefully instruct each student involved throughout their career and propel Waterloo Rocketry to even greater heights as future students become involved.

6.2 Current/Future Impact of Iterative Design

While the impacts of iterative design on LOTS were immense, the impact of this project continues through the future of the team. For the 2024 rocket, this system is already yielding benefits such as: proving airbrakes can be located ahead of the rocket CG early in the design cycle, providing a data-based decision for key engine values in an entirely new engine, and many more to come. This system should remain in use until something more effective is discovered.

6.3 Suggestions for Future Development

While many hours were spent on development for LOTS it is true that scope can be infinitely stretched to fill as many hours as students have. What follows are suggestions for future development of the iterative design approach described in this paper at the highest level:

- Refine communication as the team grows even further, both in data collection and reporting key simulation results, consider various mediums to increase clarity and reach.
- Consider how to ensure design progress is even among subsystems. Uneven progress leads to frustration and poor design decisions made with limited data. However, this should be achieved without limiting students' freedom to develop their own systems.
- Continue to refine simulation accuracy through using new tools and using previous flight data to improve processes.

- Continue to expand members view of each project to consider how it affects the entire rocket system. A rocket is an excellent learning tool to explore how small changes can have incredible impact in interdependent engineering systems.
- Continue to stress that in any simulation, garbage in = garbage out. Work to correctly describe the relevance and irrelevance of simulation data.
- Do not take design guidelines from simulation results in this paper and from other sources for granted. Due to the nature of rocketry many results are not directly transferable from one rocket to another. Consider that something good for one rocket can be bad for another.
- Many other lessons learned have been passed down through meeting notes, other documentation, and through advising the next generation of rocket designers on Waterloo Rocketry.

References

- [1] M. L. a. P. H. Benoit Dequick, "Sensitivity Analysis of a TwoPhase CFD Simulation of a 1 kN Paraffin-Fueled Hybrid Rocket Motor," 2021. [Online]. Available: <https://www.mdpi.com/1996-1073/14/20/6794>.
- [2] Dymont, Delainey Et. Al., "Team 139 Project Technical Report for the 2022 IREC," Waterloo Rocketry, 2022.
- [3] Aung, Mabel et. al., "Team 145 Project Technical Report to the 2023 Spaceport America Cup," Waterloo Rocketry, Waterloo, 2023.
- [4] H. Li, "Center of Gravity Simulation for a Hybrid Rocket Engine," Waterloo, 2016.
- [5] A. Sotnikov, "The Design of a Testable Hybrid Rocket Engine Using Software Analysis," Richmond Hill, 2023.
- [6] S. Niskanen, "OpenRocket technical documentation," 10 05 2013. [Online]. Available: <https://openrocket.info/documentation.html>. [Accessed 2024].
- [7] R. Lacau, "An Introduction To Tactical Missile Aerodynamics," 1987. [Online]. Available: <https://apps.dtic.mil/sti/tr/pdf/ADA199172.pdf>. [Accessed 2024].
- [8] J. Godard, "Fin Can Design," Waterloo, 2023.
- [9] A. Leszkowiat, "A Practical Method For Reducing Skin Friction Drag in Sounding Rockets," Oakville, ON, 2023.
- [10] Visual Crossing Corperation, "Historical Weather Data for Truth or Consequences," Accessed 2023.

Appendix 1: LOTS Mass from Conceptual Design to Flight

As reducing and tracking mass was a major focus on this project, looking at the mass and CG of each subsection of the rocket and the entire rocket together can provide insights into the process. As stated earlier the rocket was split into 5 major sections, these sections can be seen in Figure 14. Lengths in the below figure are not representative of LOTS.

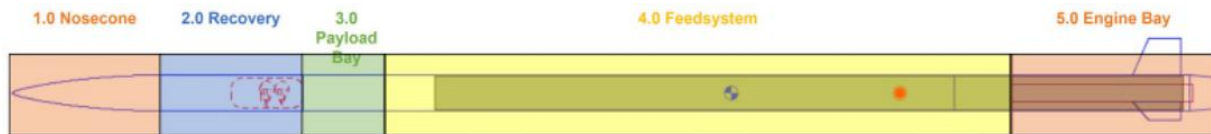


Figure 14: LOTS Sections

Note that naturally throughout the design cycle more and more components on the rocket were being recorded. This includes everything from paint (only recorded in cycle 6), bolts (often only recorded when respective fabrication was completed) and many other components. Thus, each of the graphs below is slightly false as the mass tracking in earlier cycles does not fully account for the mass required at the time. Accounting for this effect, a flat line in the below mass charts actually means a practical reduction in weight was achieved, as early cycles missed components.

Starting with the nosecone section looking at Figure 15 it is clear mass was reduced throughout the design cycle. It is also interesting to see the minimum and maximum values slowly converge as the design becomes finalized. The large drop around Cycle 2.1 is partially due to a redistribution of parts from the nosecone to the recovery section.

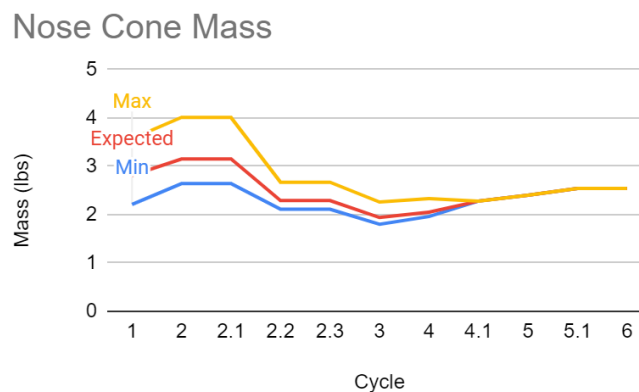


Figure 15: Nosecone Mass Throughout Design

Looking at the recovery section in Figure 16 it also shows mass decreasing throughout the design cycle. Much of this improvement was due to optimizing the weight of the recovery system. However, a large amount of this decrease is not the movement of mass but reassigning all parts physically mounted on the coupler between the recovery section and the payload section to the payload section. This made mass measurements easier but makes these charts less easy to understand.

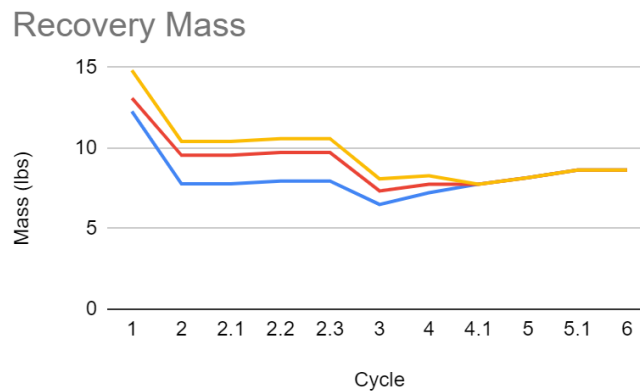


Figure 16: Recovery Mass Throughout Design

Moving onto the payload bay mass shown in Figure 17, mass of this section increased throughout the design cycle. The largest increase was due to the switch to COTS body tubes from Cycle 2.3 to Cycle 3. As mentioned previously, there was also mass reassigned to this section to make it easier to measure. Partially due to this, the maximum mass predicted at the beginning of the design cycle is below the final weight which is not a good result.

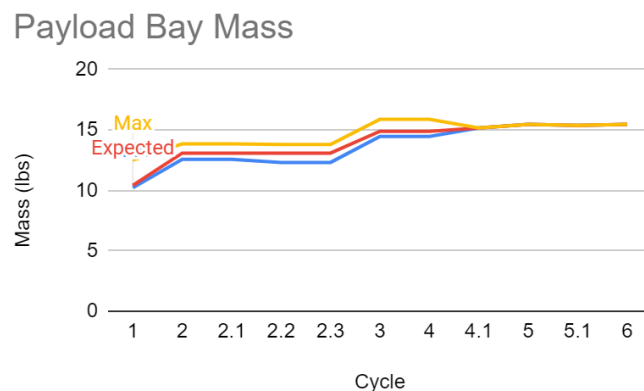


Figure 17: Payload Mass Throughout Design

Looking at the Feedsystem section graph in Figure 18, the multitude of design options near the beginning of the year shows up as a large range from minimum to maximum value. This is due to the uncertainty in improving engine efficiency. If the engine efficiency were not significantly improved, then the tanks would have been required to be longer and much heavier. However, as the engine performance improvement succeeded, the tanks could be much shorter and lighter. The lightest possible prediction at the beginning of the design cycle was a 7075 oxidizer tank which also increases the early range of values. The similarity of the final result to the initial expected prediction is uncannily close.

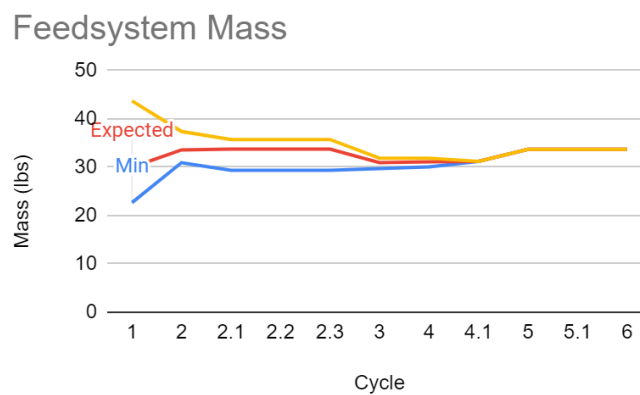


Figure 18: Feedsystem Mass Throughout Design

The mass of the engine bay shown in Figure 19 also tells a very similar story to the Feedsystem section. Initial high range of values due to engine uncertainty and then steadily converging values resulting in almost expected mass. The large drop at cycle 4 is artificial and is caused by unrecorded data.

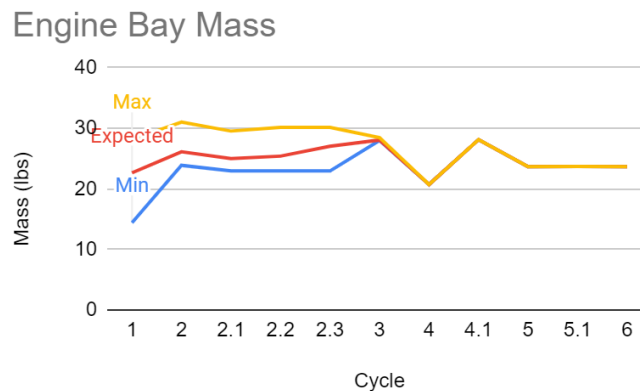


Figure 19: Engine Bay Mass Throughout Design

Combining these results, and accounting for fuel and oxidizer, the chart shown in Figure 20 can be generated. This chart shows the overall reduction of wet rocket mass throughout the project. This is one of the clearest indications of success. While it appears that mass savings are not incredibly significant, as mentioned previously, more and more components were accounted for throughout the project. This means a flat line on this chart would mean mass reduction was completed.

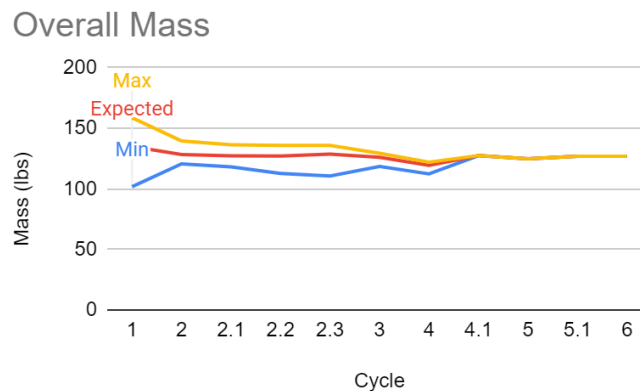


Figure 20: Overall Rocket Wet Mass Throughout Design

Another interesting chart is tracking how the center of gravity of the rocket moved throughout the design cycle. This is shown in Figure 21. While each individual section saw significant variations in center of gravity these changes tended to average out meaning the total center of gravity did not move significantly. The only major change easy to see is the dip and corresponding rise at cycle 4. This is unfortunately due to the same missing data which causes a dip in Figure 19 at cycle 4 and does not represent anything real.

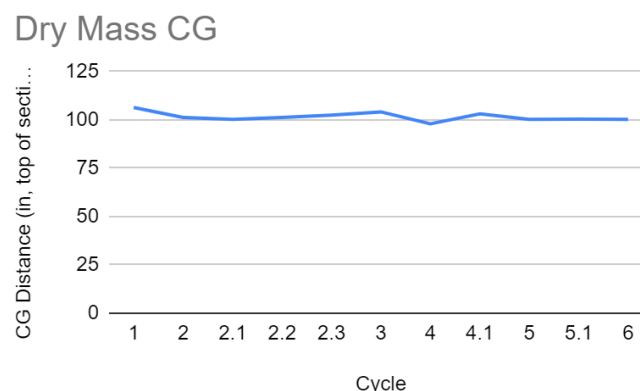


Figure 21: Dry Rocket CG Throughout Design

Note that each of these charts was generated by weighing each individual part or assembly and summing these values. The entire rocket as a whole was not practical to weigh with precision high enough to be meaningful. This creates potential for inaccuracy in these measurements. Without doing detailed analysis, this inaccuracy in dry mass at cycle 6 is expected to be around ± 0.1 lbs or less.

Special thanks to Luca Scavone for generating the above graphs and assisting with data analysis.

Appendix 2: LOTS Predicted Apogee from Conceptual Design to Flight

Another helpful chart to see the progress throughout the design of LOTS is graphing the predicted apogee at each cycle. This can be seen in Figure 22. The starting point for the graph is the projected apogee of KOTS. When starting the physical design of LOTS some obvious changes were made leading to the first major jump moving into Cycle 1. At the beginning of Cycle 2 the apogee remained fairly similar, however the fin shape optimization was implemented in Cycle 2.3. At this point, another significant improvement in apogee can be seen. Cycle 3 shows very similar results to Cycle 2.3 however in Cycle 4 the surface finish improvements were included, leading to the first simulation results consistently above 30 000ft. Cycle 5 continues to show increasing apogee due to projects that were not finished in Cycle 4 prior to learning about overshooting apogee. Cycle 6 includes reducing the oxidizer loaded into the tank to target exactly 30 000ft but is slightly above 30 000ft due to launch day weather conditions.

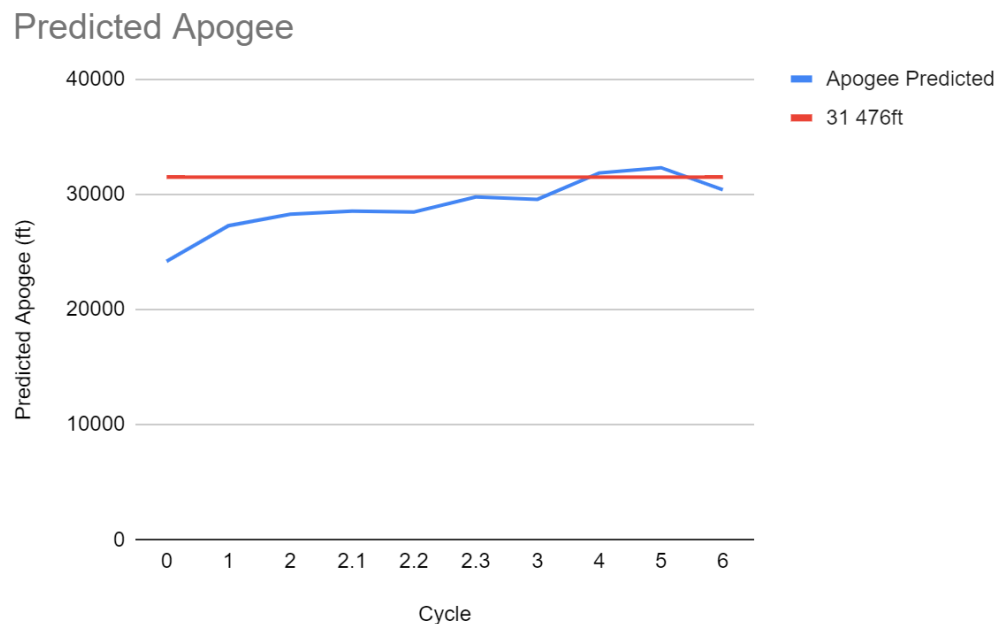


Figure 22: Predicted Apogee Throughout Design Cycle

Appendix 3: LOTS Flight Statistics

Table 5 shows some of the key LOTS flight statistics. This table was created using data gathered during flight and simulation data where physical data was not collected.

Table 5: LOTS Flight Statistics

Parameter	Value
Apogee	31 476ft
Max Mach	Mach 1.5
0-60MPH Time	0.22 Seconds
Dry Mass	84.1lbs
Wet Mass	127lbs
Length	176in
Diameter	6in
Peak Thrust	10.9kN
Off-Rail Velocity	148ft/s
Off-Rail Stability	1.79 Cal
Peak Acceleration on take-off	16.5G
Center of Gravity (from nosecone, wet)	104in
Center of Pressure (from nosecone, generic)	131in
Prop Mass Fraction (likely high estimate)	33.5%